

Job Posting Validity Checker

Sec 2025

Technical Documentation

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Overview

In 2025, Canada's labour market continues to show signs of strain. The national unemployment rate has risen to 6.9% as of October 2025, according to Statistics Canada, reflecting slow job growth and increasing economic uncertainty. At the same time, many employers report difficulty finding strong candidates due to mass applications, vague or incomplete job descriptions, and the unregulated use of AI in hiring, which contributes to confusion and mistrust among job seekers.

To address these issues and improve hiring transparency, the Government of Ontario will introduce new job-posting standards in 2026 under the Working for Workers Four Act, 2024 (S.O. 2024, c.3 – Bill 149). According to the competition's official Problem Brief, these regulations require employers to provide a realistic and legally compliant salary range, disclose any use of AI in the hiring process, clearly indicate whether the posting represents an actual vacancy, and prohibit the requirement for Canadian work experience.

These reforms aim to create a more transparent, ethical, and accessible hiring environment for both employers and job seekers.

1. Product overview

The product, JobValidation, is a tool being developed for the SEC 2025 project to address issues in Canada's labour market, such as vague job descriptions and the unregulated use of AI in hiring. It aims to improve hiring transparency in anticipation of the Government of Ontario's new job-posting standards under the *Working for Workers Four Act, 2024*.

The JobValidation is a self-contained module for data analysis, structured around a single input page featuring nine distinct input fields and an Action Button to initiate processing; its output is a structured table of analytical results where all values are presented as percentages, providing a comprehensive four-way segmentation based on four specific analytical criteria.

2. Components and Parts

1. Front-End: The user interface for interaction.
2. Backend: Processes user requests and handles the job checking logic.
3. Models (Classifier - Machine Learning Models): Used to distinguish or classify the jobs.

2.1. List of Components

- Front-end: Developed using React and Vite, with components designed using shadcn/ui.
- Back-end: Utilizes FastAPI to implement a simple Model-View-Controller (MVC) architecture.
- Machine Learning Models: Trained with scikit-learn.

Frontend Components (Shadcn UI):

- Input

- Dropdown Menu
- Button
- Textarea

Backend/API Development:

- FastAPI

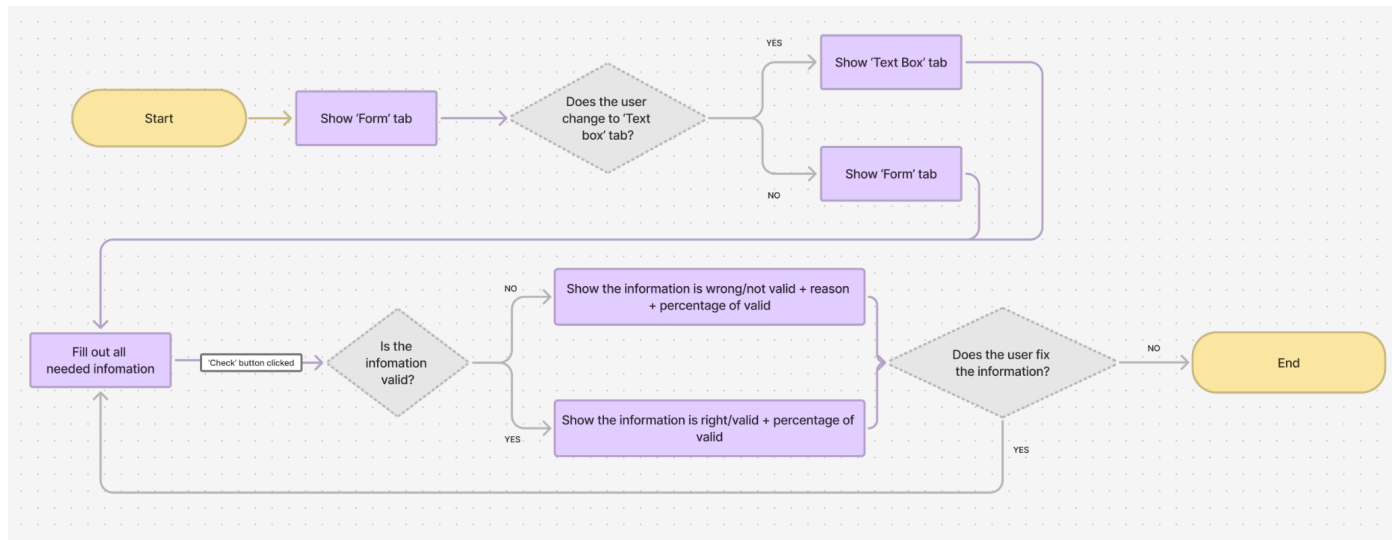
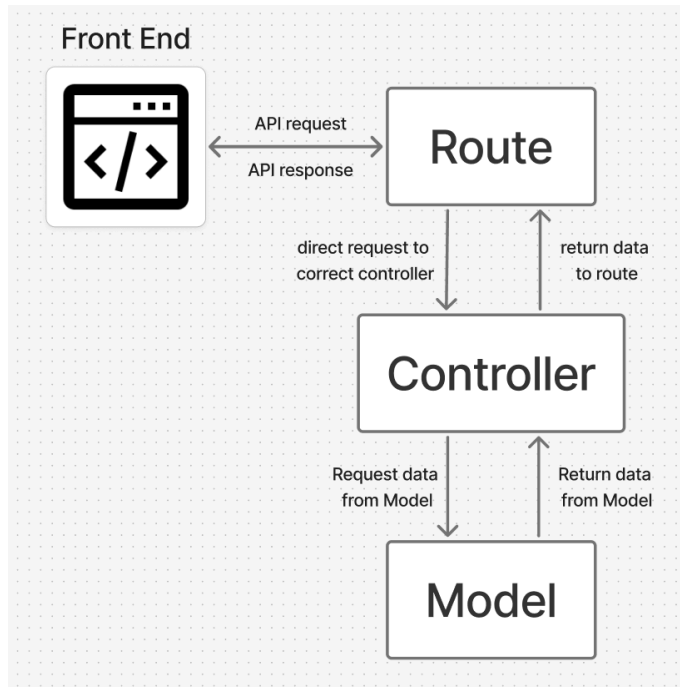
Machine Learning/Data Science:

- Scikit-learn

Core Frontend Library:

- React

2.2. Diagram



3. Setup Instructions

A) Clone

Git clone <https://github.com/minsikpaul92/sec-2025.git>

Cd front-end && npm i && npm run dev

Cd backend-end && python3 -m venv venv && source ./venv/bin/activate && fastapi dev [main.py](#)

Access <http://localhost:5173/> to use

B) Docker

Git clone <https://github.com/minsikpaul92/sec-2025.git>

Docker compose up -d

Access <http://localhost:5000/>

4. Operating Instructions

A) Dedicate website

User instruction:

Input into these text fields

- 1) Job title: Text, the title of the job
- 2) Salary range: Drop down, Pay range yearly
- 3) Location: Drop down, Location of the job
- 4) Employer: Text, name of the employer
- 5) Job Description: Text, the details description of the job
- 6) Requirements: Text, the job neede
- 7) Benefits: Text, Benefits that user receive
- 8) Employment Type: Drop down, employe type
- 9) AI is used: Drop Down

Select Result

Output: Percentage of the job transparence,

5. Technical Specifications

Support Windows 10/11, Linux with Nodejs and npm install, MacOS

6. Notes

This is a MVP product, required more development for production ready

Additional Information

- Shadcn Input: <https://ui.shadcn.com/docs/components/input>
- Shadcn Dropdown-menu: <https://ui.shadcn.com/docs/components/dropdown-menu>
- Shadcn Button: <https://ui.shadcn.com/docs/components/button>
- Shadcn Textarea: <https://ui.shadcn.com/docs/components/textarea>
- FastAPI: <https://fastapi.tiangolo.com/>
- Scikit-learn: <https://scikit-learn.org/stable/>
- React: <https://react.dev/>